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F78AB
Actuarial and Financial Mathematics

Chapter 12: Duration, Convexity and Immunisation

LEADERS IN IDEAS AND SOLUTIONS

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12.1 Duration

Consider a set of discrete future cash flows C_{t_k} to be paid at time t_k, $0 < t_1 < \ldots < t_n < T$ and a continuous cashflow at rate of payment $\rho(t)$ for $t \in [0, T]$.

The net present value of these cash flows at constant force of interest δ is

$$\text{NPV}(\delta) = \sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt.$$

What happens if the force of interest suddenly changes - how will the NPV change? Will it increase or decrease it value? By how much will it increase or decrease? We introduce the concept of *duration* to estimate the latter. It measures the sensitivity of the NPV of a set of cashflows to changes in the interest rate.

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Duration - definition

Definition 12.1 The *(Macaulay) duration* $\tau(\delta)$ of these cashflows at the force of interest δ is

$$\begin{aligned}\tau(\delta) &= -\frac{1}{NPV(\delta)} \times \frac{d}{d\delta} NPV(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k C_{t_k} e^{-\delta t_k} + \int_0^T t\rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}.\end{aligned}$$

Remarks:

(i) The duration is also called the *discounted mean term (DMT)*.

(ii) It is the weighted average time until payment of a set of cashflows, with the weights being the present value of each cashflow.

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Use of duration

$\tau(\delta)$ measures the proportionate change in $\text{NPV}(\delta)$ due to a small change in the force of interest δ.

$$\tau(\delta) \approx -\frac{1}{\text{NPV}(\delta)} \frac{\Delta \text{NPV}(\delta)}{\Delta \delta} \qquad \Rightarrow \qquad \frac{\Delta \text{NPV}(\delta)}{\text{NPV}(\delta)} \approx -\Delta\delta \cdot \tau(\delta).$$

Thus, if the force of interest changes from δ_0 to δ_1 then:

$$\frac{\text{NPV}(\delta_1) - \text{NPV}(\delta_0)}{\text{NPV}(\delta_0)} \approx -(\delta_1 - \delta_0) \cdot \tau(\delta_0)$$

This can be a good approximation for *immediate small* changes in δ.

Aside: Duration calculated as

$$-\frac{1}{\text{NPV}(i)} \times \frac{d}{di} \text{NPV}(i)$$

is called the *modified duration*. It is not equal to the Macaulay duration; can you find the relationship between them? We consider only the Macaulay duration in this course.

Zero Coupon Bonds (ZCB)

- An *n-year zero-coupon bond* is a bond which pays only a redemption amount in *n* years time, and no coupons.

- If we buy £1 nominal of an *n-year zero-coupon bond,* then we receive £1 in n years' time.

- EG if $3 nominal of a 3-year zero-coupon bond costs £0.95, then an investor can pay £0.95 today and receive a payment of £1 in exactly 3 years' time. We can work out that the gross-yield on that bond is $\left(\frac{1.0}{0.95}\right)^{1/3} - 1 = 0.017245$ or 1.7245% p.a..

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Duration of Zero Coupon Bonds

Example 12.1.3 Show that the duration of an n-year zero-coupon bond (ZCB) is n years, i.e. it does not depend on the chosen force of interest.

Solution: Duration at the force of interest δ, i.e. $\tau(\delta)$ is given by the formula:

$$\tau(\delta) = -\frac{1}{NPV(\delta)} \times \frac{d}{d\delta} NPV(\delta)$$

where the net present value of the n-year ZCB is, $NPV(\delta) = 1 \times e^{-\delta n}$.
The derivative of NPV(δ) with respect to δ is

$$\frac{d}{d\delta} NPV(\delta) = -ne^{-\delta n},$$

hence,

$$\tau(\delta) = -\frac{1}{e^{-\delta n}} \times (-ne^{-\delta n}) = n.$$

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Duration examples from notes

Exercise 12.1 Show that the duration of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{(Ia)_{\overline{n}|}}{a_{\overline{n}|}}.$$

Exercise 12.2 Show that the duration of an n-year annuity of £1 p.a. paid continuously is

$$\frac{(I\bar{a})_{\overline{n}|}}{\bar{a}_{\overline{n}|}} = \frac{(\overline{Ia})_{\overline{n}|}}{\bar{a}_{\overline{n}|}}.$$

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Duration examples

Example 12.1.4 Two investments providing the following two sets of cashflows

Time (years)	Cashflow 1	Cashflow 2
1	50	6.7
2	25	13.4
3	12	26.8
4	6	53.6

Calculate the NPV and the duration of the two sets of cashflows at $\delta = 0.05$.

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Things to consider for example 12.1.4

- Duration is the weighted mean term of the cashflows, so must lie between 1 and 4 because all the payments are between time 1 and 4.
- Duration of cashflow 1 is less than duration of cashflow 2 because cashflow 1 provides higher payments at the beginning while cashflow 2 provides higher payments towards the end.
- This means that:
  a) The weighted mean term of cashflow 1 must be smaller than the weighted mean term of cashflow 2; and
  b) Cashflow 2 is more sensitive to a change in interest rate than cashflow 1 because we receive the money later than in cashflow 1

Duration of Cashflows 1 and 2

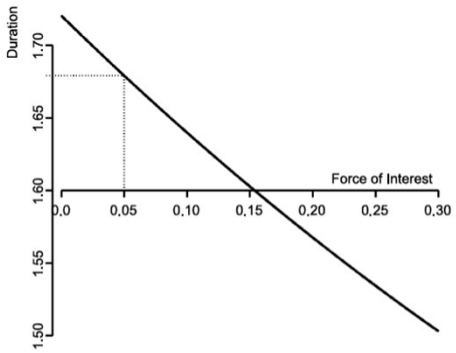


Figure 12.1: Duration of cashflow 1 in Example 12.1.4

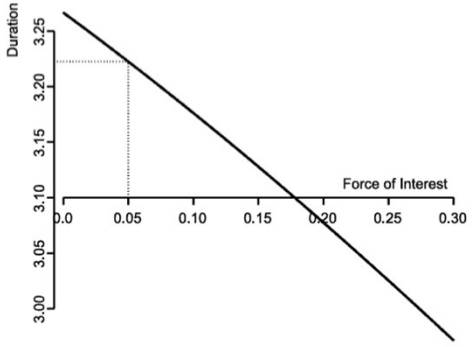


Figure 12.2: Duration of cashflow 2 in Example 12.1.4

For both, the sensitivity of the NPV of cashflows decreases with an increasing force of interest.
Duration (mean term) is decreasing with an increasing force of interest, since payments far in the future become less important with the market fore of interest is high.
This can be compared to the DPP for the 2 cashflows, which is increasing if the market force of interest is increasing.

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Example 12.1.5

Example 12.1.5 Consider the follong fixed-income securities.

A: term 10 years, coupon 12% p.a. paid annually in arrears, redemption at par.

B: term 25 years, coupon 3% p.a. paid annually in arrears, redemption 120%.

$$\begin{aligned}\tau_A(\delta) &= \frac{1\times 12\times v + 2\times 12\times v^2 + \cdots + 10\times 12\times v^{10} + 10\times 100\times v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12\left(v + 2v^2 + \cdots + 10v^{10}\right) + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12(Ia)_{\overline{10}|} + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}}\end{aligned}$$

Example 12.1.5 cont…

Similarly,

$$\tau_{\text{B}}(\delta) = \frac{3(Ia)_{\overline{25}|} + 120 \times 25 \times v^{25}}{3a_{\overline{25}|} + 120v^{25}}.$$

Assume the current force of interest is 4% p.a.

δ	$\text{NPV}_{\text{A}}(\delta)$	$\text{NPV}_{\text{B}}(\delta)$	$\tau_{\text{A}}(\delta)$	$\tau_{\text{B}}(\delta)$
4%	163.97	90.61	7.1451	17.7975

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12.1.5 cont – if the interest rate changes

Sudden shift in force of interest	Actual price change A	Actual price change B	Estimated price change $\approx -(\delta_1 - \delta_0)\tau(0.04)$ A	Estimated price change $\approx -(\delta_1 - \delta_0)\tau(0.04)$ B
From 4% to 3%	+7.465%	+19.92%	+7.1451%	+17.7975%
From 4% to 5%	-6.843%	-15.98%	-7.1451%	-17.7975%

For the actual price change, we used the following figures:

δ	$\text{NPV}_\text{A}(\delta)$	$\text{NPV}_\text{B}(\delta)$
3%	176.21	108.66
5%	152.75	76.13

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Convexity

Definition 12.3 For any set of discrete cashflows $\{(C_{t_k}, t_k); k = 1, 2, \ldots, n\}$ and a continuous cashflow at rate of payment $\rho(t)$ for all $t \in [0, T]$, we define the *convexity* of the combined cashflows as

$$\begin{aligned} CONV(\delta) &= \frac{1}{NPV(\delta)} \frac{d}{d\delta^2} NPV(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k^2 C_{t_k} e^{-\delta t_k} + \int_0^T t^2 \rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}. \end{aligned}$$

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Convexity

Exercise 12.3 Show that the convexity of an n-year zero-coupon bond (ZCB) is n^2 years, i.e. it does not depend on the chosen force of interest.

Solution:

$$
\begin{aligned}
NPV(\delta) &= e^{-\delta n} \\
\frac{d}{d\delta} NPV(\delta) &= -ne^{-\delta n} \\
\frac{d}{d\delta^2} NPV(\delta) &= n^2 e^{-\delta n} \\
\implies CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2} NPV(\delta) \\
&= \frac{n^2 e^{-\delta n}}{e^{-\delta n}} \\
&= n^2
\end{aligned}
$$

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Convexity examples

Exercise 12.4 Show that the convexity of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{\sum_{k=1}^{n} k^2 v^k}{a_{\overline{n}|}}.$$

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Solution Exercise 12.4

$$
\begin{aligned}
NPV(\delta) &= \sum_{k=1}^{n} e^{-\delta k} \\
\frac{d}{d\delta^2} NPV(\delta) &= \sum_{k=1}^{n} k^2 e^{-\delta k} \\
\Longrightarrow CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2} NPV(\delta) \\
&= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{\sum_{k=1}^{n} e^{-\delta k}} \\
&= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{\sum_{k=1}^{n} v^k} \\
&= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{a_{\overline{n}|}}
\end{aligned}
$$

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Exercise 12.5 Show that the convexity of an n-year annuity of £1 p.a. paid continuously is

$$\frac{2(\bar{I}\bar{a})_{\overline{n}|} - n^2 v^n}{\delta \bar{a}_{\overline{n}|}}.$$

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Solution Exercise 12.5

$$
\begin{aligned}
NPV(\delta) &= \int_0^n e^{-\delta t}\,dt \\
\frac{d}{d\delta^2}NPV(\delta) &= \int_0^n t^2 e^{-\delta t}\,dt \\
\Longrightarrow CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2}NPV(\delta) \\
&= \frac{1}{\int_0^n e^{-\delta t}\,dt}\int_0^n t^2 e^{-\delta t}\,dt \\
&= \frac{1}{\bar{a}_{\overline{n}|}}\int_0^n t^2 e^{-\delta t}\,dt.
\end{aligned}
$$

where (using integration by parts),

$$
\begin{aligned}
\int_0^n t^2 e^{-\delta t}\,dt &= -\frac{n^2}{\delta}e^{-\delta n} + \frac{2}{\delta}\int_0^n te^{-\delta t}\,dt \\
&= -\frac{n^2}{\delta}e^{-\delta n} + \frac{2}{\delta}(\overline{Ia})_{\overline{n}|} \\
\Longrightarrow CONV(\delta) &= \frac{1}{\bar{a}_{\overline{n}|}}\left(\frac{2}{\delta}(\overline{Ia})_{\overline{n}|} - \frac{n^2}{\delta}e^{-\delta n})\right) \\
&= \frac{2(\overline{Ia})_{\overline{n}|} - n^2v^2}{\delta\bar{a}_{\overline{n}|}}.
\end{aligned}
$$

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11.3 Immunisation

Set-up: financial institution (e.g. life insurance company or pension fund). The institution has future net liability cashflows of L_{t_k} at time t_k, for $k = 1, 2, \ldots$. It holds assets which provide cashflows A_{s_n} at time s_n, for $n = 1, 2, \ldots$. Let the current force of interest be δ_0 p.a. We assume that

$$NPV_A(\delta_0) = \sum_{n=1}^{\infty} A_{s_n} e^{-\delta_0 s_n} = NPV_L(\delta_0) = \sum_{k=1}^{\infty} L_{t_k} e^{-\delta_0 t_k}. \qquad (12.4)$$

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What if interest rates change?

Since the NPV is a function of the force of interest, equation (12.4) may no longer hold if the force of interest changes. Hence, the institution faces the following problem.

Problem: How to choose assets so that an instantaneous change in the force of interest from δ_0 to δ_1 will not result in $NPV_A(\delta_1) < NPV_L(\delta_1)$?

"Perfect Matching": Choose $s_1 = t_1$, $s_2 = t_2$, ..., and then $A_{t_k} = L_{t_k}$ for all k (not usually a practical solution).

In cases in which "Perfect Matching" is not possible it might still be possible for the institution to choose their assets such that a small immediate change in the force of interest will not result in insufficient NPV of the assets. This provides at least some protection against the risk associated with a change in the interest rate.

The following theorem shows how a financial institution can protect itself against immediate small changes in the force of interest. This protection is called immunisation.

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Redington's theory of immunisation

Suppose that the market force of interest is δ_0 p.a. If the following conditions hold:

$$NPV_A(\delta_0) = NPV_L(\delta_0), \qquad \text{"NPV matching"} \tag{12.5}$$

$$\tau_A(\delta_0) = \tau_L(\delta_0), \qquad \text{"Duration matching"} \tag{12.6}$$

$$CONV_A(\delta_0) > CONV_L(\delta_0), \qquad \text{"Convexity condition"} \tag{12.7}$$

then there exists $\epsilon > 0$ such that for all δ with $|\delta - \delta_0| < \epsilon$ the following holds

$$NPV_A(\delta) \geq NPV_L(\delta),$$

with equality holding only at $\delta = \delta_0$.

Interpretation: If the conditions (12.5), (12.6) and (12.7) are fulfilled, then an *immediate, small* change in the force of interest from δ_0 to δ_1 will give:

$$NPV_A(\delta_1) > NPV_L(\delta_1).$$

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Redington's theory of immunisation

Define $f(\delta) = NPV_A(\delta) - NPV_L(\delta)$.

$$\text{Equation (12.5)} \quad \Rightarrow \quad f(\delta_0) = 0.$$

$$\begin{aligned} \text{Equations (12.5) and (12.6)} \quad &\Rightarrow \quad NPV'_A(\delta_0) - NPV'_L(\delta_0) = 0 \\ &\Rightarrow \quad f'(\delta_0) = 0 \\ &\Rightarrow \quad f \text{ has a turning point at } \delta_0. \end{aligned}$$

$$\begin{aligned} \text{Equations (12.5) and (12.7)} \quad &\Rightarrow \quad NPV''_A(\delta_0) - NPV''_L(\delta_0) > 0 \\ &\Rightarrow \quad f''(\delta_0) > 0 \\ &\Rightarrow \quad f \text{ has a minimum point at } \delta_0. \end{aligned}$$

It follows that (i) $f(\delta)$ has a local minimum at $\delta = \delta_0$, and (ii) $f(\delta_0) = 0$.
Hence for all δ close to δ_0,

$$f(\delta) > f(\delta_0) = 0.$$

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Redington's theory of immunisation

Remarks:

1. If conditions (12.5),(12.6) and (12.7) hold, the financial institution is said be immunized (in the sense of Redington) against immediate small changes in the interest rate.

2. The financial institution is immunized against immediate small changes in the interest rate assuming a flat yield curve. (Generalizes to immediate parallel shifts of a yield curve.)

3. Suppose (12.5) and (12.6) hold and there are only discrete cashflows. Then (12.7) is equivalent to

$$\sum_{n=1}^{\infty}(s_n - \tau_A(\delta_0))^2 A_{s_n} \exp(-\delta_0 s_n) > \sum_{k=1}^{\infty}(t_k - \tau_L(\delta_0))^2 L_{t_k} \exp(-\delta_0 t_k) \qquad (12.8)$$

i.e. the "spread of the assets" around the common τ is greater than the "spread of the liabilities". A similar result holds for continuous cashflows.

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Immunisation example

Example 12.3.1 A life office has to pay £1 million in 10 years time. The effective rate of interest is $i = 4\%$ p.a.. The available securities are a 5-year and a 15-year zero-coupon bond (ZCB).
Calculate how much of each ZCB the life office should buy so that it is immunised in the sense of Redington.

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Immunisation examples cont…

Exercise 12.8 Show that with the solution above we have:

(i) If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ +£870.

(ii) If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£703.

Hint: Use the nominal amounts of the ZCBs to calculate their prices under the new interest rate.

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Immunisation examples cont…

Exercise 12.9 Show that if all money had been invested in the 5-year ZCB:

(i) If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ -£35 093.

(ii) If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£30 088.

Hence we see that immunisation works against profits as well as losses.

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Immunisation final example

Exercise 12.10 An insurance company has issued a number of policies for which a total liability of £1 000 000 is payable in exactly 10 years' time. The insurance company holds sufficient funds to cover the liability on the basis of a force of interest of δ per annum and intends to invest this money solely in the purchase of a 5-year and a 20-year zero-coupon bond (ZCB).

What proportion of the company's funds should be invested in the 5-year ZCB to ensure that the discounted mean term of the assets equals the discounted mean term of the liabilities at a constant force of interest of δ per annum given that this is also the market basis for pricing all stocks?

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Attendance matters!

